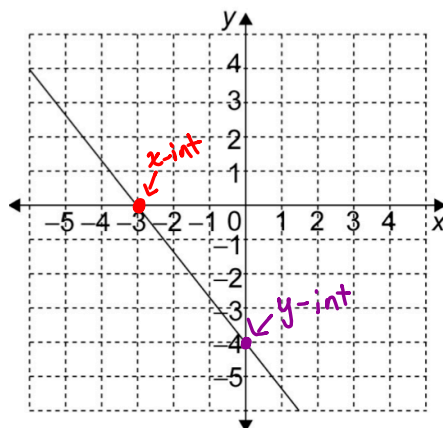


6.3 Graphing Using Intercepts

Part 1: Do It Now!

What are the x and y intercepts of the following line:



x -intercept: $(-3, 0)$

y -intercept: $(0, -4)$

When a line is written in standard form, $Ax + By + C = 0$, or the form $Ax + By = -C$, it is easy to graph the line using

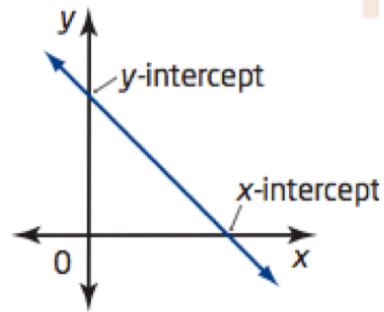
x and y -intercepts.

The **x-intercept** is the x -coordinate of the point where the line crosses the x -axis.

At the x -intercept, $y = 0$.

The **y-intercept** is the y -coordinate of the point where the line crosses the y -axis.

At the y -intercept, $x = 0$.



Example 1:

Determine the intercepts for the line $2x - 3y - 6 = 0$ and use these points to graph the line.

To find the x -intercept,
set $y = 0$ and solve:

$$2x - 3(0) - 6 = 0$$

$$2x - 6 = 0$$

$$\frac{2x}{2} = \frac{6}{2}$$

$$x = \frac{6}{2}$$

$$x = 3$$

$$x\text{-int} : (3, 0)$$

To find the y -intercept,
set $x = 0$ and solve:

$$2(0) - 3y - 6 = 0$$

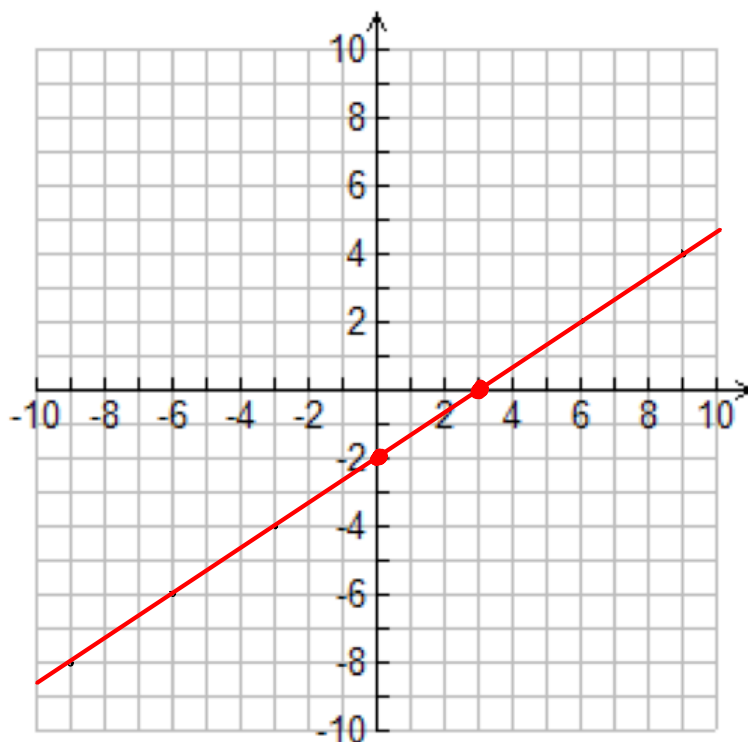
$$-3y - 6 = 0$$

$$\frac{-3y}{-3} = \frac{6}{-3}$$

$$y = \frac{6}{-3}$$

$$y = -2$$

$$y\text{-int} : (0, -2)$$



Example 2:

Determine the intercepts for the line $2x - y = 7$ and use these points to graph the line.

To find the x -intercept,
set $y=0$ and solve:

$$2x - 0 = 7$$

$$2x = 7$$

$$x = \frac{7}{2}$$

$$x = 3.5$$

$$x\text{-int: } (3.5, 0)$$

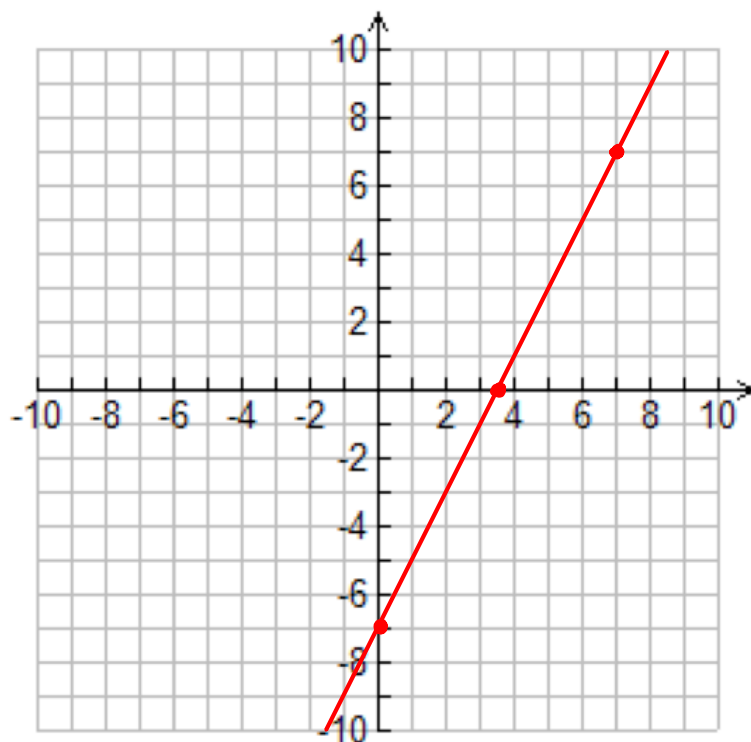
To find the y -intercept,
set $x = 0$ and solve:

$$2(0) - y = 7$$

$$-y = 7$$

$$y = -7$$

$$y\text{-int: } (0, -7)$$



Example 3:

a) Determine the intercepts for the line $5x - 6y + 30 = 0$.

x-int

$$5x - 6(0) + 30 = 0$$

$$5x + 30 = 0$$

$$5x = -30$$

$$x = \frac{-30}{5}$$

$$x = -6$$

y-int

$$5(0) - 6y + 30 = 0$$

$$-6y + 30 = 0$$

$$-6y = -30$$

$$y = \frac{-30}{-6}$$

$$y = 5$$

b) Use the intercepts to determine the slope of the line.

Point 1: $(-6, 0)$

Point 2: $(0, 5)$

$$m = \frac{5 - 0}{0 - (-6)} = \frac{5}{6}$$

Remember:

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

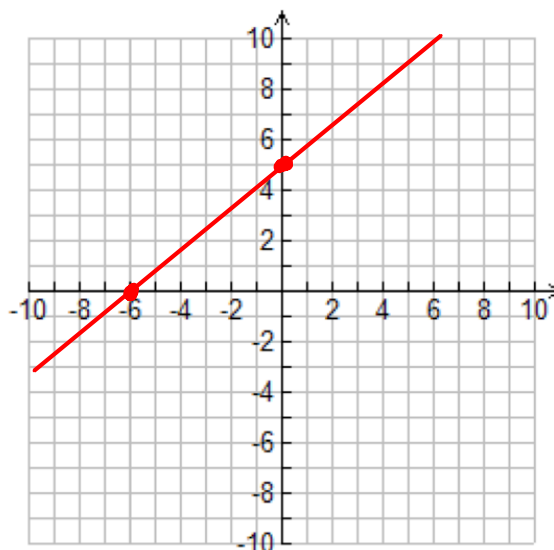
c) Write the equation of the line

$$m = \frac{5}{6}$$

$$b = 5$$

$$y = \frac{5}{6}x + 5$$

d) Graph the line



Example 4: Determine the slope of the line whose x -intercept is -4 and y -intercept is -6.

$$\text{Point 1: } (\overset{x_1}{-4}, \overset{y_1}{0})$$

$$\text{Point 2: } (\underset{x_2}{0}, \underset{y_2}{-6})$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-6 - 0}{0 - (-4)} = \frac{-6}{4} = -\frac{3}{2}$$

Consolidate:

State the steps needed to graph a line using the intercepts.

- 1) Solve for the y -intercept by setting $x = 0$
- 2) Solve for the x -intercept by setting $y = 0$
- 3) Plot the intercepts and draw a straight line through them